

TITLE_NEW_PAPER

*Omar Iskandarani**

Independent Researcher, Groningen, The Netherlands

April 19, 2026

Abstract

Abstract text here

Keywords: PAPER_KEYWORDS

0.1 Helical-Locking Universality Across Distinct Mechanical Architectures

A central observation from physical prototypes is that two mechanically distinct systems can realize closely similar collective motion when driven by a single central helix:

1. a rigid monolithic triple-gear assembly printed in PLA, in which the triple gear and helical axle are fixed-shape contact elements;
2. a modular trefoil-based carrier printed as multiple compliant TPU segments, later folded into a closed trefoil-like body, together with a separate printed central helical insert.

These systems are not topologically identical and are not mechanically identical. Nevertheless, both appear to realize the same effective three-sector progression around a central helical driver. This motivates a distinction between:

$$\text{topology,} \quad \text{mechanical architecture,} \quad \text{helical-locking class.} \quad (1)$$

Definition. Let ϕ denote the phase of the central helical driver, and let θ_i ($i = 1, 2, 3$) denote the phases of the three surrounding sectors. A system belongs to a given *helical-locking class* if its coarse-grained motion obeys

$$\theta_i = \sigma_i \kappa_i \phi + \delta_i, \quad i = 1, 2, 3, \quad (2)$$

with fixed handedness signs $\sigma_i \in \{+1, -1\}$, effective pitch ratios κ_i , and phase offsets δ_i , and if no phase discontinuity (“gear jumping”) occurs over the admissible motion range.

Architecture-independent equivalence. Two systems A and B are said to be *helical-lock equivalent*,

$$A \sim_{\text{HL}} B, \quad (3)$$

if they satisfy:

$$\theta_i^{(A)}(\phi) - \sigma_i^{(A)} \kappa_i^{(A)} \phi \approx \delta_i^{(A)}, \quad (4)$$

$$\theta_i^{(B)}(\phi) - \sigma_i^{(B)} \kappa_i^{(B)} \phi \approx \delta_i^{(B)}, \quad (5)$$

with small residuals, and if the two systems share approximately the same tuple

$$\mathcal{C}_{\text{HL}} = \{(\sigma_i, \kappa_i, \delta_i)\}_{i=1}^3. \quad (6)$$

~~This definition does not require identical topology or identical material realization.~~

* Email: info@omariskandarani.com

ORCID: 0009-0006-1686-3961

DOI: 10.5281/zenodo.xxx

Effective theory. At coarse scale, both rigid and compliant realizations reduce to

$$\mathcal{L}_{\text{eff}} = \frac{I_{\text{eff}}}{2} \dot{\phi}^2 - V_{\text{eff}}(\phi), \quad (7)$$

where

$$I_{\text{eff}} = I_{\phi} + \sum_{i=1}^3 I_i \kappa_i^2 + I_{\text{comp}}, \quad (8)$$

and I_{comp} is an architecture-dependent correction term encoding elastic compliance, local deformation, and contact redistribution.

For a rigid PLA realization,

$$I_{\text{comp}} \approx 0 \quad (9)$$

to leading order.

For a folded TPU realization,

$$I_{\text{comp}} \neq 0, \quad (10)$$

but if the phase-locking law (2) remains intact, then the compliant architecture belongs to the same helical-locking class as the rigid one.

Interpretation. This suggests that the robust invariant is not the detailed body plan itself, but the existence of:

$$\text{three sectors} + \text{single central helix} + \text{continuous non-jumping phase transfer}. \quad (11)$$

SST implication. Within SST, this motivates the conjecture that particle-like identity may be determined first by a *locking universality class*, and only secondarily by precise topological representative or material realization. In this view, distinct carriers can implement the same effective degree of freedom if they preserve the same locked helical transport law.

Empirical status. The claim is experimentally motivated: printed PLA and TPU realizations with distinct architectures nevertheless appear to exhibit similar collective motion under a central helical drive. The present evidence is kinematic and qualitative; quantitative validation requires measured phase residuals, torque curves, and slip thresholds.

Minimal falsifiers. The conjecture fails if:

1. the phase-lock law differs between the two realizations;
2. one system exhibits unavoidable phase slip while the other remains locked;
3. the effective periodic potential $V_{\text{eff}}(\phi)$ differs qualitatively;
4. the inferred collective mode spectrum is not asymptotically equivalent.

Measured locking ratio. For the printed triple-gear realization, direct manual rotation shows that five complete turns of the central helix produce one complete rotation of each of the three linked rings. Hence the experimentally inferred locking coefficient is

$$\kappa = \frac{1}{5}, \quad (12)$$

so that

$$\theta_i = \sigma_i \frac{1}{5} \phi + \delta_i, \quad i = 1, 2, 3. \quad (13)$$

This provides a concrete measured representative of the helical-locking law.